Vortex-Sheet Representation of Leading-Edge Vortex Shedding from Finite Wings

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A characteristic feature of flows past many oscillating airfoils and wings is the leading-edge vortex (LEV). Although considerable progress has been made in the low-order modeling of LEV formation over airfoils, the prediction of LEV formation on finite wings remains a challenge. A low-order method is presented for modeling the formation of the LEV on finite wings. An unsteady vortex-lattice method has been augmented to model the effect of the LEV, which is modeled as a vortex sheet. The criticality of leading-edge suction is used to modulate the formation of the LEV at different spanwise locations of the wing. This feature enables prediction of the spanwise location of the LEV onset and the subsequent nonuniform growth of the LEV on the wing. A good comparison of the predicted flow features is seen with results from a higher-order Reynolds-averaged Navier–Stokes code. The predicted forces are also in reasonable agreement. The LEV sheet model has the potential to be used in a variety of low-order methods for rapid prediction of unsteady finite-wing aerodynamics.

Nomenclature

A_n	=	time-dependent Fourier coefficients for thin airfoil theory
a	=	influence coefficient, 1/m
b	=	wingspan, m
C_L	=	lift coefficient
C_M	=	moment coefficient about wing root quarter-chord
C_N	=	normal-force coefficient
C_S	=	leading-edge suction force coefficient
c	=	airfoil chord, m
$\boldsymbol{F}_N$	=	normal component of force, N
$\boldsymbol{F}_S$	=	leading-edge suction force, N
i	=	chordwise panel (cell) index
$i_{\max}$	=	number of panels (cells) along the chord
K	=	reduced pitch rate; $\dot{\alpha}c/2U_\infty$
$\mathrm{LESP}_{\mathrm{crit}}$	=	critical value of leading-edge suction parameter
N	=	total number of vortex rings
n	=	number of time steps
$\boldsymbol{n}$	=	normal vector on surface, m
p	=	pressure, N/m^2
p_∞	=	freestream static pressure, N/m^2
Re	=	Reynolds number based on average chord length
r	=	radial coordinate, m
$\boldsymbol{r}$	=	Cartesian vector, m
r_c	=	cutoff radius (vortex-core radius, singularity radius), m
$\boldsymbol{r}_e$	=	Cartesian vector of end point of a vortex filament, m
$\boldsymbol{r}_s$	=	Cartesian vector of starting point of a vortex filament, m
S	=	wing planform area, m^2
t	=	time, s
t^*	=	nondimensional time; $U_\infty t/c$
U_∞	=	magnitude of freestream velocity, m/s
$\boldsymbol{v}$	=	velocity vector, m/s
$\boldsymbol{v}_m$	=	velocity vector induced by motion, m/s
$\boldsymbol{v}_r$	=	velocity vector for rollup, m/s
$\boldsymbol{v}_\infty$	=	freestream velocity vector, m/s
x	=	chordwise coordinate from the leading edge, m
y	=	spanwise coordinate from wing root, m
α	=	angle of attack, deg
$\dot{\alpha}$	=	rate of change of angle of attack, rad/s
Γ	=	circulation, m^2/s
γ	=	wing bound vorticity, m/s
$\boldsymbol{\gamma}$	=	wing bound vorticity vector, m/s
γ_x	=	x component of wing bound vorticity, m/s
γ_y	=	y component of wing bound vorticity, m/s
Δp	=	pressure difference, N/m^2
ΔS	=	area of a discretized cell, m^2
Δt	=	time step, s
Δx	=	chordwise panel size, m
$\Delta\boldsymbol{x}$	=	rollup displacement vector, m
Δx_W	=	distance of first shed vortex ring from trailing edge, m
Δy	=	spanwise panel size, m
θ	=	chordwise angular coordinate; $x \equiv (c/2)(1 - \cos\theta)$
ρ	=	density, kg/m^3
ϕ	=	velocity potential, m^2/s

Subscripts

b	=	wing bound circulation
L	=	leading-edge vortex
l	=	on lower surface
u	=	on upper surface
W	=	trailing-edge vortex and wake
τ	=	tangential component
1	=	at forwardmost bound-vortex ring

I. Introduction

LEADING-EDGE vortices (LEVs), which are vortices shed from the leading edges of wings, can be observed in nature (in swimming and flying animals [1–3], and in autorotating seeds [4]) and in engineering (in rotorcraft [5–7], wind turbines [8], turbochargers [9], swept and delta wings [10,11], and micro air vehicles [12]). The presence of an LEV plays a significant role in aerodynamic loads. The LEV structure has a low-pressure region in its core, which augments the lift if the LEV is on the suction side of the wing surface [13]. In addition, advection of the LEV causes a

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variation in the pitching moment of the wing [14]. Thus, the understanding of the aerodynamics of LEV formation and propagation is of interest for real-world engineering applications.

The main engineering concerns for studies on the LEV are the conditions of its initiation and its subsequent development. Many investigations have revealed the connection between the onset of LEV formation and factors such as leading-edge radius [15,16], Reynolds number [5,15,16], and motion kinematics [6,17–31]. Experimental and computational observations of three-dimensional LEV phenomena report intriguing flow patterns like Ω-shaped vortical structures [25,32–36]. Moreover, computational and experimental studies have quantified the effects of LEV growth and detachment on aerodynamic forces and moments [27,28,37]. Rotational effects on LEV development on rotor blades and flapping wings have also been studied [4,38].

An inviscid and incompressible fluid flow theory can be used to develop a low-order method to study the characteristics of LEV formation. This assumption allows fluid flow to be described by potential flow theory with viscous effects modeled using bound vortices and shear layers, discretized as a series of point vortices (two-dimensional discrete-vortex method) or a series of vortex rings (three-dimensional vortex-sheet representation). There have been several successful models describing LEV formation in two-dimensional flows using conformal mapping methods [39–41] and using discrete-vortex methods [42–44]. However, the existing literature relating to theoretical studies of three-dimensional LEV formation model is more limited. One of the popular LEV formation models for finite wings is for delta wings. Most LEV models for delta wings postulate a pair of steady conical vortex structures shed from the leading edges [45,46]. This assumption is unsuitable for application to general wing geometries in unsteady motion [7]. Furthermore, earlier models of LEV formation on finite wings have idealized the LEV structure as a collection of vortex filaments instead of a vortex sheet. For example, Brown and Michael [47] proposed to use a single vortex filament representing the core of a LEV for a delta wing system. Kandil et al. [48] extended this single-core model to multiple vortex filaments shed from the leading edge of a delta wing in place of LEV sheets. Katz [49] assumed a large vortex ring extending from the leading edge to the trailing edge in which the front edge of the ring behaves like a single LEV filament and the aft edge is a single trailing-edge vortex (TEV) filament. These examples of three-dimensional LEV models give good insights, but their scope is limited to problems in which the expected LEV structures can be assumed to be two-dimensional. Moreover, the determination of LEV onset is also a complicated problem to model in the sense that the leading edge is mathematically a singular point in thin airfoil theory and that flow separation cannot be determined by a vanishing-wall-shear criterion under unsteady flow conditions [50]. For these reasons, conventional LEV formation models often assume continuous shedding [41,43,49] or set an ad hoc trigger condition [51] for LEV formation. The use of theoretical models providing a condition for LEV initiation, as typified by the leading-edge suction parameter (LESP) concept proposed by Ramesh et al. [44], has been limited to two-dimensional flows.

In this paper, a new LEV formation model for finite-wing motion in unsteady flow is developed and incorporated in an unsteady vortex-lattice method (UVLM). Although the development of the first steady vortex-lattice method (VLM) dates back to Hedman's work in 1965 [52], the VLM and UVLM, with various modifications and enhancements, are used even today for low-order modeling and engineering applications, with recent examples ranging from flight dynamics analysis [53,54], analysis of yacht sails [55], calculation of aerodynamic interference effects [56–58], poststall analysis [59,60], flapping-wing analysis [61–63], wind turbines [64,65], design optimization [66,63], and aeroelasticity [67–69]. The objective of the current work is to augment the UVLM to model the primary LEV structure using a vortex-sheet representation; to simulate the initiation, formation, and convection of the LEV; and to calculate its influence on aerodynamic loads. It is known that the primary LEV is often accompanied by a secondary LEV [70] and sometimes by smaller subordinate LEVs [71]. The modeling of multiple LEVs on a three-dimensional wing using vortex sheets is a difficult problem. It can be argued that, at least for the initial evolution of the LEV sheet, the influence of the secondary and smaller LEVs on the aerodynamics loads is significantly less than that of the primary LEV sheet [10,48]. For this reason, the present LEV formation model focuses only on the primary vortex sheet. An extension of the LESP concept proposed by Ramesh et al. [44,72] to a finite-wing system and an approach for modeling discontinuous vortex shedding within the vortex-sheet representation are presented. The performance of the new LEV model is verified by comparing predictions with those obtained using computational fluid dynamics (CFD) simulations, and the computational savings afforded by the use of the low-order theoretical model are also quantified.

II. Theoretical Model of Leading-Edge Vortex Formation

This section presents the new modeling aspects of the vortex-sheet representation of LEV shedding on finite wings. To avoid redundancy, aspects of the formulation that are common to conventional unsteady vortex-lattice methods ([73] pp. 419–433) are abbreviated. A more detailed derivation may be found in Ref. [74].

An overview of the vortex-sheet representation using vortex rings is first presented in Sec. II.A. Calculation of the shed vortex ring for the trailing-edge vortex sheet is then reviewed in Sec II.B. The new approach for modeling the leading-edge vortex sheet is then described: first, by discussion of the critical leading-edge suction for initiation of LEV shedding in Sec. II.C; and next, by discussion of the shedding of vortex rings for the LEV sheet in Sec. II.D. The subsequent subsections discuss the solution of the linear system, the calculation of the aerodynamic loads, and the vortex-sheet rollup. Finally, a summary of the entire procedure is presented in Sec. II.H using a flowchart.

A. Overview of the Vortex-Sheet Representation

A vortex-sheet representation is an aerodynamic [73] model for the vorticity distribution within the boundary layer and wake based on potential flow theory. In this model, a thin-volume vorticity distribution is lumped into a vorticity distribution on a thin sheet. This lumped vorticity distribution on the sheet is discretized into a number of vortex filaments ([75] pp. 23–46) and is further redefined as a series of closed vortex rings ([73] pp. 419–433), as shown in Fig. 1. As done in a traditional unsteady vortex-lattice method ([73] pp. 419–433), the bound vorticity on the wing is modeled as a bound-vortex sheet using vortex rings distributed along the mean camber line. The trailing-edge vortex sheet models the vorticity in the shear layer emanating from the wing trailing edge. The bound-vortex and trailing-edge vortex sheets are modeled similar to the approach used in a traditional UVLM ([73] pp. 419–433). The novel contribution of this work is the leading-edge vortex sheet that is used to model vorticity that emanates from the rounded leading edge subsequent to initiation of the LEV formation. The LEV sheet is also discretized using vortex rings. The trailing-edge and leading-edge vortex sheets are both free vortex sheets, and they are allowed to deform by advection of the vortex rings with the local velocity. These and other details are presented in the following subsections. In the current work, each tip vortex is modeled as a vortex filament (an edge of the tip ring) that is attached to the wingtip edge. In reality, vorticity separates from the edges and rolls up into conical tip-vortex sheets. Implementation of such separated-tip-vortex sheets in conjunction with LEV sheets resulted in numerical problems when the sheets intersected with each other. For this reason, the current method is only applicable to wings with an aspect ratio (AR) of four and higher, for which earlier work [76] has shown that there is little difference in the predicted forces from the attached-tip-vortex and the separated-tip-vortex models.

The vortex ring is described by considering it as a small flat plate. According to thin airfoil theory, the representative bound circulation and zero-normal-flow control point must be, respectively, defined at the quarter-chord location and at the three-quarter-chord location in the chordwise direction and at the center in the spanwise direction.

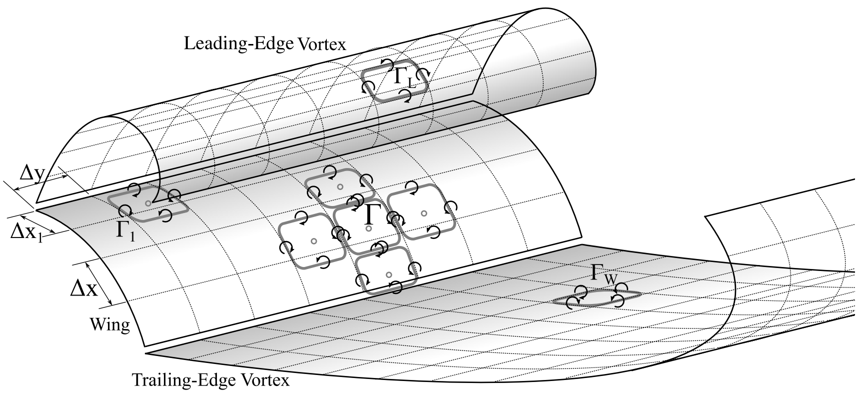


Fig. 1 Vortex-sheet representation and discretization.

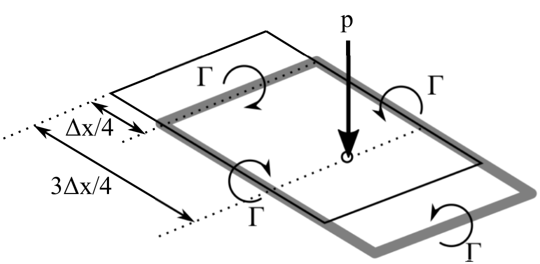


Fig. 2 Wing element (thin black line) and vortex-ring element (thick gray line).

Thus, a bound-vortex ring for a correspondingly discretized wing element is positioned as shown in Fig. 2. It is noteworthy that the addition of two vortex rings adjacent to one another determines the strength of the circulation distribution on their border line. The vortex-ring representation is able to reduce the cumbersome mathematical treatment of vorticity distribution to simple algebra, as seen in subsequent sections. In the vortex-sheet representation, the velocity field is obtained by integrating velocity components induced by each set of lumped vortex rings.

The choice of vortex rings for the discretization of vortex sheets was made in this work for both the relative simplicity of the implementation and for keeping computational costs low. Future implementations would benefit from the use of higher-order discretization and alternate formulations for the surface vorticity, which have advantages in robustness when handling wake–surface and wake–wake interactions. Some examples of these higher-order formulations are the use of curved panels with continuous vorticity distributions [77], the use of distributed vorticity elements for modeling wakes [78], the use of enriched basis functions [79], the discretization of vortex sheets using segments that maintain the curvature [80], and the recent surface vorticity panel formulation [81,82].

B. Trailing-Edge Vortex Sheet

The modeling of the trailing-edge vortex sheet, which is similar to the approach used in the traditional UVLM ([73] pp. 419–433), is reviewed in this subsection. In this discussion, it is assumed that there is no leading-edge vortex sheet. The trailing-edge vortex sheet uses vortex rings to model the vorticity in the shear layer that is shed from the trailing edge of the wing. To explain the calculation of the strength of the TEV rings, it is useful to consider a few bound-vortex and TEV rings on a strip of the wing in the vicinity of the trailing edge as shown in Fig. 3.

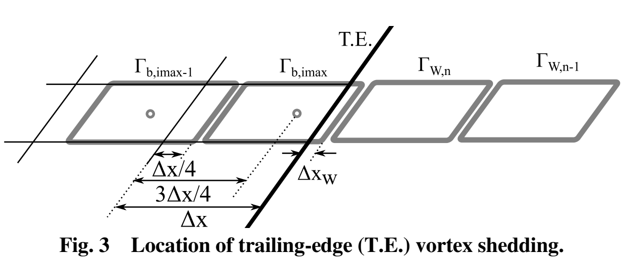


Fig. 3 Location of trailing-edge (T.E.) vortex shedding.

Referring to Fig. 3, the distance of the forward edge of the latest TEV ring from the trailing edge of the wing is denoted by Δx_W. Following the recommendation of Katz and Plotkin ([73] pp. 419–433), the current work uses $\Delta x_W = 0.3 U_\infty \Delta t$. For the UVLM strip of the wing shown in Fig. 3, using the symbol $i\,\max$ to denote the number of vortex rings in the chordwise direction of the wing, the vortex strength of the forward edge of the latest TEV ring is $(\Gamma_W^n - \Gamma_{b,i\max}^n)$, where Γ_W^n is the strength of the latest TEV ring (shed at the nth time step) and $\Gamma_{b,i\max}^n$ is the strength of the aftmost bound-vortex ring at the nth time step. For any UVLM strip, the vortex strengths of the spanwise-oriented bound-vortex legs from the leading edge to the trailing edge are $\Gamma_{b,1}$, $(\Gamma_{b,2} - \Gamma_{b,1})$, $(\Gamma_{b,3} - \Gamma_{b,2})$, and so on, with the vortex strength of the aftmost leg (seen in Fig. 3) being $(\Gamma_{b,i\max} - \Gamma_{b,i\max-1})$. Thus, the total bound circulation over the entire chord for this strip is the sum of these strengths, which is simply the strength of the aftmost bound-vortex ring $\Gamma_{b,i\max}$. Therefore, the change in the strip's bound-vortex-sheet strength between the previous and current time steps is $(\Gamma_{b,i\max}^n - \Gamma_{b,i\max}^{n-1})$.

To satisfy the Kelvin condition $(\mathrm{D}\Gamma/\mathrm{D}t = 0)$, for each UVLM strip, the total change in the strengths of the bound-vortex and TEV sheets between the current and previous time steps (n and $n-1$) is set to zero as follows:

$$\left(\Gamma_{b,i\max}^n - \Gamma_{b,i\max}^{n-1}\right) + \left(\Gamma_W^n - \Gamma_{b,i\max}^n\right) = 0 \tag{1}$$

which simplifies to the following:

$$\Gamma_W^n = \Gamma_{b,i\max}^{n-1} \tag{2}$$

In the time-stepping approach, the latest TEV ring and all previous TEV rings convect downstream with the local velocity, as will be described later in Sec. II.G, with no change in the strength. Thus, as seen in Fig. 3, the strength of the second TEV ring is Γ_W^{n-1}, that of the third TEV ring is Γ_W^{n-2}, and so on.

C. Leading-Edge Suction Parameter

The leading-edge suction parameter, proposed by Ramesh et al. [44,72], is an aerodynamic parameter that provides a nondimensional measure of the suction at the leading edge of the airfoil. Criticality of the LESP has been successfully used as a criterion in low-order modeling for initiation of LEV formation on unsteady airfoils [44]. In this low-order modeling approach, initiation of LEV formation on airfoils occurs when the instantaneous LESP value exceeds the critical LESP. The instantaneous LESP can be determined using the unsteady thin-airfoil theory, as will explained in the following paragraphs of this subsection; the determination of the critical LESP, or $\mathrm{LESP_{crit}}$, is briefly discussed here. As shown by Ramesh et al. [44,83], the $\mathrm{LESP_{crit}}$ for a given airfoil and chord Reynolds number is largely independent of motion kinematics for the range of motions in which LEV formation is not accompanied by significant trailing-edge separation. Such

LEV-dominated flows without significant trailing-edge separation are characteristic of airfoils and wings undergoing high-rate pitch or plunge oscillations [44]. It is well understood that LEV formation at low subsonic Mach numbers depends on pressure gradients and viscous effects at the leading edge; it is therefore affected by the chord Reynolds number and airfoil geometry parameters, such as the leading-edge radius and the thickness-to-chord ratio. In the LESP approach, the dependence of the LEV formation on these factors is captured by determining the $\mathrm{LESP_{crit}}$ from CFD or experimental results for each airfoil and Reynolds number of interest. The main advantage of the LESP concept is that, once the $\mathrm{LESP_{crit}}$ value is determined for one motion, low-order prediction of LEV formation for any other motion is possible for that airfoil geometry and Reynolds number combination. More details and validation of this idea for unsteady airfoils were provided by Ramesh et al. [44]. Salient information is presented here with an emphasis on extension of the idea to finite wings modeled using the UVLM.

The instantaneous LESP value for an unsteady airfoil is determined using an unsteady thin-airfoil theory. The chordwise distribution of bound circulation on an airfoil is expressed as a Fourier series in a classical thin-airfoil theory ([84] pp. 88–91). Following the unsteady thin-airfoil theory of Ref. ([73] pp. 387–399), the chordwise distribution of bound circulation can be expressed as a function of the angular coordinate as follows:

$$\gamma(\theta,t) = 2U_\infty\left[A_0(t)\frac{1+\cos\theta}{\sin\theta} + \sum_{n=1}^{\infty} A_n(t)\sin(n\theta)\right] \tag{3}$$

Here, θ is an angular coordinate that relates to the chordwise coordinate x as $x = c(1-\cos\theta)/2$.

Ramesh et al. [44] observed that the determining factor for the leading-edge suction for an airfoil is the circulation at the leading edge $\gamma(0,t)$, which is represented by the first coefficient $A_0(t)$. The value of this coefficient at any time instant is defined as the instantaneous LESP [44]. In the low-order method described by Ramesh et al. [44], called the LDVM or LESP-modulated Discrete Vortex Method, the discrete-time implementation of the unsteady thin-airfoil theory ([73] pp. 387–399) is augmented with a discrete-vortex method for modeling the LEV shedding. When analyzing the unsteady motion of an airfoil using the LDVM [44], LEV formation is assumed to occur when the instantaneous value of the LESP exceeds the predetermined $\mathrm{LESP_{crit}}$ value for that airfoil and Reynolds number. At each time step when the LEV formation is active, a discrete vortex is released at the leading edge. The strength of this discrete vortex is such that it brings the LESP value back to the critical value. All these discrete vortices are convected with the local flow velocity at each time step. When the LESP decreases below the critical value, LEV shedding is terminated. Thus, the LDVM is able to simulate flow past unsteady airfoils with intermittent LEV shedding, and it serves as a motivation for the current work on prediction of LEV formation on finite wings.

It is worth mentioning that, although the leading-edge suction parameter term used here is reminiscent of Polhamus's leading-edge suction analogy [85,86], there is not much connection between the two. In Polhamus's analogy, which is applied to swept slender wings with a stationary leading-edge vortex, the leading-edge suction force calculated for the section is rotated by 90 deg to account for the high lift from the stationary LEV. The LESP idea, on the other hand, is used to identify the onset of LEV formation and its subsequent evolution and termination.

Because the current effort uses the LESP to determine the initiation of LEV formation on a finite wing, an approach is needed to determine the spanwise variation of the LESP along the wing at every time step. When the bound vorticity is modeled using a chordwise distribution of vortex rings, there is no A_0 term, and an alternate approach is needed to determine the LESP. In a vortex-lattice method, it is clear that the strength of the forwardmost bound-vortex leg is connected to the leading-edge vorticity in thin-airfoil theory, and hence to the A_0 parameter. Direct use of the strength of this forwardmost vortex leg, labeled as Γ_1 in this discussion and in Fig. 1, to the LESP value is undesirable because the LESP value would then be dependent on the number of chordwise lattices used in the VLM discretization.

An approach for determining the value of the LESP in a vortex-lattice method was developed by Aggarwal [87], which is used in this effort. In this approach, it is assumed that, on a small chordwise region near the leading edge occupied by the forwardmost vortex lattice, all the terms in the Fourier-series representation [Eq. (3)], except for the A_0 term, would be negligibly small because these coefficients go to zero anyway at the leading edge in thin-airfoil theory. With this assumption, the value of Γ_1 in the vortex-lattice representation can be connected to the A_0 value in thin-airfoil theory by equating the Γ_1 value to the integrated bound-circulation strength due to the A_0 term over the chordwise extent of the forwardmost lattice (from $x = 0$ at the leading edge to the aft end of the forwardmost lattice, denoted here by Δx_1, as shown in Fig. 1). The resulting equation is as follows:

$$\begin{aligned}\Gamma_1(t) &= \int_0^{\Delta x_1}\gamma(x,t)\,\mathrm{d}x = \int_0^{\cos^{-1}(1-(2\Delta x_1/c))}\gamma(\theta,t)\frac{c}{2}\sin\theta\,\mathrm{d}\theta\\ &= \int_0^{\cos^{-1}(1-(2\Delta x_1/c))}\left\{2U_\infty A_0(t)\frac{1+\cos\theta}{\sin\theta}\right\}\frac{c}{2}\sin\theta\,\mathrm{d}\theta\\ &= cU_\infty A_0(t)\left[\cos^{-1}\left(1-\frac{2\Delta x_1}{c}\right)+\sin\left\{\cos^{-1}\left(1-\frac{2\Delta x_1}{c}\right)\right\}\right]\end{aligned} \tag{4}$$

The resulting expression allows the connection between the A_0 value in thin-airfoil theory with the Γ_1 value in the vortex-lattice method as follows:

$$A_0(t) = \frac{\Gamma_1(t)}{U_\infty c[\cos^{-1}(1-(2\Delta x_1/c))+\sin\{\cos^{-1}(1-(2\Delta x_1/c))\}]} \tag{5}$$

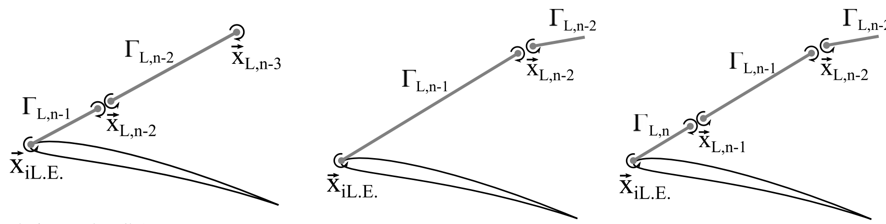
With this equivalence, the expression on the right side of Eq. (5) can be used to calculate the value of the LESP in a vortex-lattice method. Although this approach resulted in an LESP that was essentially independent of the number of chordwise lattices, the value was found to be short of the A_0 calculation in Ref. [44] by approximately 13%. To account for the differences between the two methods and to enable use of the same $\mathrm{LESP_{crit}}$ in the two methods, Eq. (5) was modified by using a scaling factor of 1.13 on the right side, resulting in Eq. (6). As shown in Ref. [87], this expression for A_0, calculated from a discrete-vortex-lattice approach, resulted in A_0 values that were invariant with the number of vortex lattices, and that consistently matched up with the A_0 values predicted by the method of Ref. [44] for a range of airfoil motions:

$$A_0(t) = \frac{1.13\Gamma_1(t)}{U_\infty c[\cos^{-1}(1-(2\Delta x_1/c))+\sin\{\cos^{-1}(1-(2\Delta x_1/c))\}]} \tag{6}$$

In the current implementation of the UVLM, the $\mathrm{LESP}(y,t)$ for each strip, located at spanwise coordinate y, is calculated by equating it to the $A_0(t)$ from Eq. (6), which is calculated using the Γ_1 for that strip at that time step in the calculation.

D. Leading-Edge Vortex Sheet

The leading-edge vortex sheet models the vorticity shed from the leading edge using vortex rings. The current approach extends the LEV shedding implemented in the unsteady airfoil work of Ramesh et al. [44] to determine the strength of the LEV ring shed from each strip of the wing. Although a TEV ring is shed from the trailing edge at each time step, an LEV ring for any strip of the UVLM is shed only when $\mathrm{LESP}(t) > \mathrm{LESP_{crit}}$ for that strip. When $\mathrm{LESP}(t) > \mathrm{LESP_{crit}}$, the strength of the LEV ring is determined so that the LESP is brought back to the predetermined $\mathrm{LESP_{crit}}$ value. In the current implementation, because the strength of the frontmost bound-vortex



a) At $t = (n-1)\Delta t$, before LEV convection

b) At $t = n\Delta t$, before LEV convection

c) At $t = n\Delta t$, after LEV calculation, before LEV convection

Fig. 4 Placement of leading-edge (L.E.) vortex ring.

leg Γ_1 is connected to the instantaneous LESP value through Eq. (6), constraining the instantaneous LESP to the critical value essentially sets a constraint on the maximum value of Γ_1. This subsection first presents the methodology for the placement of the LEV ring and then discusses the implementation of the Kelvin condition.

At each time step, to position the new LEV ring, the current unsteady vortex-lattice method uses the method proposed by Ansari et al. [43]. The idea, illustrated in Fig. 4 using a slice of the LEV sheet near the leading edge, places the new LEV at one-third of the distance from the leading edge to the location of the last LEV. This method is depicted in Fig. 4 and is expressed as follows:

$$\boldsymbol{x}_{L,n-1} = \frac{2}{3}\boldsymbol{x}_{i,\mathrm{L.E.}} + \frac{1}{3}\boldsymbol{x}_{L,n-2} \tag{7}$$

This formulation is advantageous in the sense that it can simultaneously take into account the wing motion and vortex convection. When the LEV sheet is represented using vortex rings, as shown in Figs. 4b and 4c, the placement of a new LEV ring is implemented by splitting a previously shed LEV ring into two LEV rings at the point determined by Eq. (7). Then, as shown in Fig. 4c, the nascent LEV ring of strength $\Gamma_{L,n}$ is defined as the vortex ring lying across the leading edge and $\boldsymbol{x}_{L,n-1}$. At subsequent time steps, this and all other LEV rings convect with the local velocity as described in Sec. II.G with no change in strength.

The aft end of the new nascent vortex ring lies on the geometric leading edge and results in a vortex filament of strength $\Gamma_{L,n}$ at this edge. In the UVLM formulation, however, there is no bound-vortex filament at the geometric leading edge. The frontmost bound-vortex filament is located at one-quarter of the lattice width behind the leading edge, as seen in Fig. 1. To eliminate the vortex filament at the geometric leading edge, another vortex ring, referred to here as the "pseudovortex ring," is introduced. This pseudovortex ring, of strength $\Gamma_{L,n}$, has the front edge at the geometric leading edge and the aft edge at the aft edge of the rearmost bound-vortex ring, as shown in Fig. 5. With the use of such a ring, the geometric leading edge ends up having a zero-strength vortex filament; and the strength of the spanwise-oriented vortex filament immediately in front of the leading edge is $(\Gamma_L^n - \Gamma_L^{n-1})$, where Γ_L^{n-1} is the strength of the pseudovortex ring at the previous time step $(n-1)$. At every time step when LEV shedding is active (i.e., when the instantaneous LESP is greater than the $\mathrm{LESP_{crit}}$), the strength of the pseudovortex ring Γ_L^n is set so as to make the instantaneous LESP equal to the $\mathrm{LESP_{crit}}$.

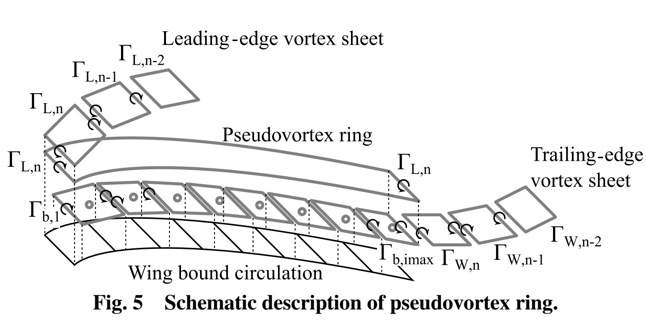


Fig. 5 Schematic description of pseudovortex ring.

It is noted that the shape of the pseudovortex ring used in the current implementation is one of several possible shapes. An alternative implementation could have used a pseudovortex ring with the aft edge coinciding with the front edge of the frontmost bound-vortex ring. Doing so would have made the strength of the frontmost bound-vortex filament equal to $(\Gamma_{b,1} - \Gamma_{L,n})$, and it would have required the appropriate alteration of the LESP calculation in Eq. (6). Instead, in the current formulation, by aligning the aft edge of the pseudovortex ring with the aft edge of the rearmost bound-vortex ring, this alteration is avoided. However, the Kelvin condition needs to be revisited.

To satisfy the Kelvin condition $(\mathrm{D}\Gamma/\mathrm{D}t = 0)$, for each UVLM strip, the total change in the strengths of the bound-vortex, LEV, and TEV sheets between the current and previous time steps (n and $n-1$) is set to zero. As discussed in Sec. II.B, the change in the section bound-circulation strength between the previous and current time steps is $(\Gamma_{b,i\max}^n - \Gamma_{b,i\max}^{n-1})$. Using the set of vortex rings illustrated in Fig. 5 as an example, the strengths of the spanwise filaments of the LEV sheet, starting from the free edge of the LEV sheet, are Γ_L^{n-2}, $(\Gamma_L^{n-1} - \Gamma_L^{n-2})$, and $(\Gamma_L^n - \Gamma_L^{n-1})$. Of these, the spanwise filament closest to the leading edge is the new leading-edge vortex generated between time steps $(n-1)$ and n. Thus, the change in the LEV circulation strengths between the previous and current time steps is $(\Gamma_L^n - \Gamma_L^{n-1})$. Because of the inclusion of the pseudo vortex ring, the strength of the new TEV filament just aft of the section trailing edge is $(\Gamma_W^n - \Gamma_{b,i\max}^n - \Gamma_L^n)$. For each strip of the UVLM, the Kelvin condition is used to determine the value of the unknown strength of the latest TEV ring at the current time step Γ_W^n by setting the sum of the changes in the bound-vortex, LEV, and TEV sheets to zero as follows:

$$\left(\Gamma_{b,i\max}^n - \Gamma_{b,i\max}^{n-1}\right) + \left(\Gamma_L^n - \Gamma_L^{n-1}\right) + \left(\Gamma_W^n - \Gamma_{b,i\max}^n - \Gamma_L^n\right) = 0 \tag{8}$$

which simplifies to the following:

$$\Gamma_W^n = \Gamma_{b,i\max}^{n-1} + \Gamma_L^{n-1} \tag{9}$$

E. Solution of Linear System

The distribution of wing bound circulation is derived by imposing the zero normal flow at the wing surface. This boundary condition can be described by Eq. (10):

$$(\boldsymbol{v}_b + \boldsymbol{v}_\infty + \boldsymbol{v}_m + \boldsymbol{v}_W + \boldsymbol{v}_L) \cdot \boldsymbol{n} = 0 \tag{10}$$

The velocity induced by motion $\boldsymbol{v}_m$ is determined by the given motion kinematics, and the velocities induced by shed TEVs and LEVs ($\boldsymbol{v}_W$ and $\boldsymbol{v}_L$) are calculated by applying the Biot–Savart law to each shed vortex ring in the flowfield. Thus, the only unknown value is $\boldsymbol{v}_b$, which is determined by the wing bound circulation Γ_b. Equation (10) is rewritten for all N_b wing bound-vortex rings and the other known vortices using influence coefficients as follows:

$$[a_b]\begin{pmatrix}\Gamma_{b,1}\\ \vdots\\ \Gamma_{b,N_b}\end{pmatrix}+\begin{pmatrix}(\boldsymbol{v}_\infty+\boldsymbol{v}_{m,1})\cdot\boldsymbol{n}_1\\ \vdots\\ (\boldsymbol{v}_\infty+\boldsymbol{v}_{m,N_b})\cdot\boldsymbol{n}_{N_b}\end{pmatrix}+[a_W]\begin{pmatrix}\Gamma_{W,1}\\ \vdots\\ \Gamma_{W,N_W}\end{pmatrix}+[a_L]\begin{pmatrix}\Gamma_{L,1}\\ \vdots\\ \Gamma_{L,N_L}\end{pmatrix}=0 \tag{11}$$

In Eq. (11), the brackets [ ] correspond to an influence coefficient (IC) matrix. This linear system is solved for the unknown distribution Γ_b.

F. Calculation of Aerodynamic Load

Aerodynamic loads on the wing are obtained from the calculated pressure distribution on its surface and the leading-edge suction force. The pressure distribution is obtained from the wing bound-circulation distribution. The suction force is evaluated using potential theory for plane flow.

The pressure distribution on the surface is derived through the application of Bernoulli's equation along the surface at the midpoint of each UVLM strip. Using the far upstream boundary condition ($p=p_\infty$, $\boldsymbol{v}=\boldsymbol{v}_\infty$, $\phi=0$) as the reference, the pressure distribution along the wing surface can be described as follows:

$$\frac{p_\infty-p}{\rho}=\frac{|\boldsymbol{v}_\tau|^2}{2}-\frac{|\boldsymbol{v}_\infty|^2}{2}+\frac{\partial\phi}{\partial t} \tag{12}$$

$$\begin{aligned}\frac{\Delta p}{\rho}&=\frac{p_l-p_u}{\rho}\\&=\left(\frac{|\boldsymbol{v}_{\tau,u}|^2}{2}+\frac{\partial\phi_u}{\partial t}\right)-\left(\frac{|\boldsymbol{v}_{\tau,l}|^2}{2}+\frac{\partial\phi_l}{\partial t}\right)\\&=(\boldsymbol{v}_\infty+\boldsymbol{v}_m+\boldsymbol{v}_W+\boldsymbol{v}_L)\cdot\left\{\gamma_y(x)\cdot\hat{x}_\tau+\gamma_x(y)\cdot\hat{y}_\tau\right\}\\&\quad+\frac{\partial(\phi_u-\phi_l)}{\partial t}\end{aligned} \tag{13}$$

For calculating the last term on the right side of Eq. (13), the expression for the velocity potentials on either side of the bound-vortex sheet, the strength of which is denoted by $\gamma(x)$, is used:

$$\frac{\partial\phi}{\partial x}(x,0\pm)=\pm\frac{\gamma(x)}{2} \tag{14}$$

For the time instants when there is no active LEV shedding, because there is no vortex sheet upstream of the leading edge, $\phi_u=\phi_l$ just upstream of the leading edge. Thus, the potential difference between the upper and lower surfaces at a location x on the wing section can be obtained by integration of the bound-vortex-sheet strength. Using a dummy variable x' to denote the distance along the chord, the integration is performed from the start of the vortex sheet at the strip leading edge ($x'=0$) to $x'=x$. Because the bound-vortex-sheet strength is represented using vortex rings, the potential difference at x is simply the strength of the vortex ring in which that x is located, which is denoted here by $\Gamma_{b,x}$:

$$\phi_u-\phi_l=\int_{x'=0}^{x}\gamma(x')\,\mathrm{d}x'=\Gamma_{b,x} \tag{15}$$

For time instants when there is LEV shedding, the vortex sheet does not start at the leading edge but instead starts at the free edge of the LEV sheet that is attached to the leading edge. For this case, the potential difference between the upper and lower surfaces at a location x on the wing section is obtained by an integration from the free edge of the LEV sheet to the leading edge, followed by integration from the leading edge to the location x. The integral of the LEV sheet strength from the start of the sheet to the leading edge is simply the strength of the latest LEV ring. Referring to Fig. 5, this integral is Γ_L^n for the nth time step. Thus, the potential difference at location x on the chord is as follows:

$$\phi_u-\phi_l=\Gamma_L^n+\Gamma_{b,x} \tag{16}$$

Therefore, the last term on the right side of Eq. (13) is as follows:

$$\frac{\partial(\phi_u-\phi_l)}{\partial t}=\frac{\Delta(\Gamma_L+\Gamma_{b,x})}{\Delta t}=\frac{(\Gamma_L^n-\Gamma_L^{n-1})+(\Gamma_{b,x}^n-\Gamma_{b,x}^{n-1})}{\Delta t} \tag{17}$$

where Δt is the time step. When LEV shedding is not active, Γ_L is zero and has no contribution.

The normal force $\boldsymbol{F}_N$ and the normal-force coefficient C_N acting on the wing element can then be evaluated by

$$\boldsymbol{F}_N=\sum\Delta p\Delta S\boldsymbol{n} \tag{18}$$

$$C_N=\frac{-F_N}{(1/2)\rho U_\infty^2 S} \tag{19}$$

As shown in Ref. [44], given the instantaneous value of $A_0(t)$, the suction force magnitude F_S can be calculated by potential flow theory:

$$F_S=\pi\rho c U_\infty^2 A_0^2 \tag{20}$$

$$C_S=\frac{F_S}{(1/2)\rho U_\infty^2 S}=\frac{\sum_y\pi\rho c U_\infty^2 A_0^2\Delta y}{(1/2)\rho U_\infty^2 S}=\frac{2\pi}{S}\sum_y cA_0^2\Delta y \tag{21}$$

The positive direction of $\boldsymbol{F}_S$ is upstream and parallel to the camber line at the leading edge. Defining an appropriate unit vector, a vector representation of the leading-edge suction force can be obtained. Using the evaluation of $A_0(t)=\mathrm{LESP}(y,t)$, the current UVLM with the new LEV formation model can predict C_L for any angle of attack, taking into account the contribution of the normal and suction force coefficients as follows:

$$C_L=C_N\cos\alpha+C_S\sin\alpha \tag{22}$$

G. Vortex-Sheet Rollup

Vortex-sheet rollup, which is a model for wake vorticity convection, is achieved by calculating the displacement of the vortex rings as driven by the local velocity field $\boldsymbol{v}_r$. The displacement of a vortex ring during a time step in an inertial frame of reference $\Delta\boldsymbol{x}$ is simply described as

$$\Delta\boldsymbol{x}=\boldsymbol{v}_r\Delta t \tag{23}$$

where $\boldsymbol{v}_r$ has induced-velocity contributions from the bound, trailing-vortex, and leading-edge vortex sheets: $\boldsymbol{v}_r=\boldsymbol{v}_b+\boldsymbol{v}_W+\boldsymbol{v}_L$. Dovgi and Shekhovtsov [88] used the following first-order approximation of Euler's method to determine the displacement vector:

$$\Delta\boldsymbol{x}=\frac{\boldsymbol{v}_r^n+\boldsymbol{v}_r^{n-1}}{2}\Delta t \tag{24}$$

Equation (24) uses information from the present and the previous time steps and is employed as the time-marching scheme for vortex rollup in the present effort. Although this scheme was chosen for its simplicity, it is noted, however, that future implementations may benefit from considering higher-order time integrations that may be more suitable for cases involving complex wing–wake and wake–wake interactions.

In calculating the induced velocities due to the vortex filaments, to avoid singularities, some viscous-effect model is required [89]. Considering the computational cost and expected accuracy, the Lamb–Oseen model [90], restricted by a singular radius r_c, is employed in this paper to desingularize the Biot–Savart law [88,91] as follows:

$$v(r) = \frac{\Gamma}{4\pi} \frac{r_s \times r_e}{|r_s \times r_e|^2} (|r_s| + |r_e|) \left(1 - \frac{r_s \cdot r_e}{|r_s||r_e|}\right) \{1 - e^{-(r/r_c)^2}\} \tag{25}$$

where r_s and r_e are the vectors from the point at which the velocity is calculated to the start and end points of the filament, respectively. The value of r_c is set to a small fraction, typically less than 0.5, of the anticipated smallest vortex-ring dimension in the problem under consideration.

H. Summary of the Theoretical Model

The procedure for the current implementation of the UVLM with a leading-edge vortex sheet is summarized using the flowchart in Fig. 6. The procedure becomes the same as that for a conventional UVLM ([73] pp. 419–433) when LEV formation is not initiated. For each strip of the ULVM, LEV rings are shed when the LESP is greater than the predetermined critical value.

III. Methodology

A. Cases

Three finite-wing geometries are considered for testing the UVLM with LEV formation. The airfoil section is the SD7003 ([92] p. 121), as shown in Fig. 7. This airfoil, designed for use in low-Reynolds-number applications, has a thickness-to-chord ratio of 8.5%, and has been used in several recent studies of low-Reynolds-numbers unsteady airfoils and wings. Figure 8 shows the three wing

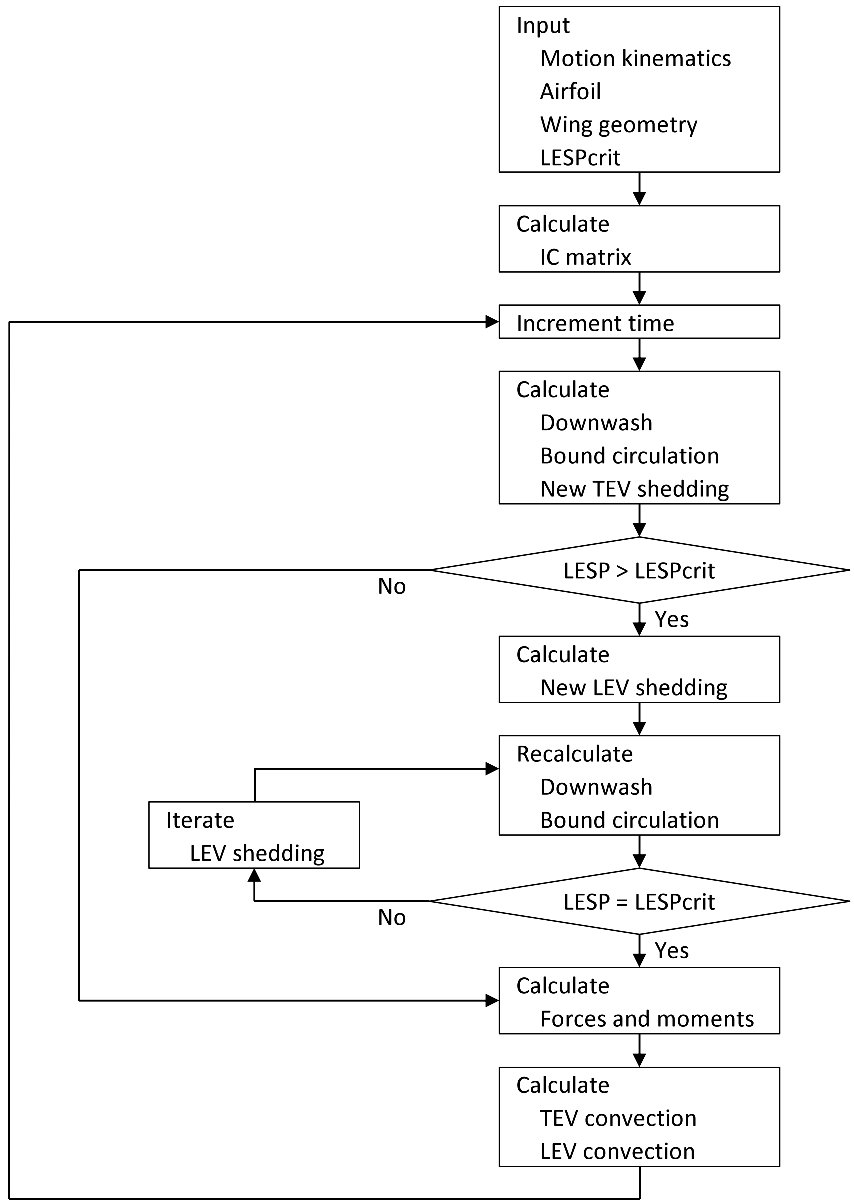


Fig. 6 Schematic flowchart.

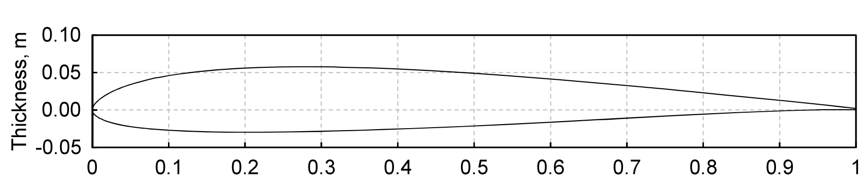
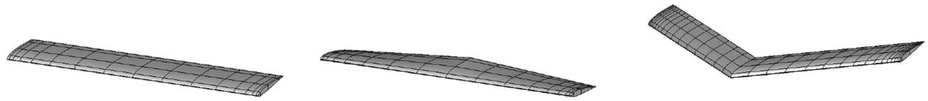


Fig. 7 SD7003 airfoil section.



a) Rectangular wing (case 1) b) Twisted wing (case 2) c) Swept wing (case 3)

Fig. 8 Geometries of the three wings used as test cases.

geometries that are tested in this paper, and Table 1 lists the details. The three cases, labeled cases 1–3, are for an AR = 6 constant-chord wing with different tip-twist angles and sweep angles, which result in different patterns of LEV formation [76]. The twisted wing has 10 deg higher incidence at the tip as compared to the root. Although constant-chord wings have been used for illustration in this work, the method can be used for analysis of arbitrary wing planforms. All the studies in this work have been performed for a chord Reynolds number of 20,000. This value was chosen because our previous studies have shown that the LESP criterion successfully predicts initiation of LEV formation on airfoils (in two-dimensional flow) at Reynolds numbers between 10,000 and 40,000. A value of $\mathrm{LESP}_{\mathrm{crit}} = 0.27$ has been used for this airfoil and Reynolds number combination. This value was obtained from the two-dimensional (2-D) CFD analysis of the SD7003 airfoil [76].

B. Motion Parameters

The current work focuses on a pure pitching motion. The wing pitch axis is placed at the quarter-chord ($x = c/4$) location of the wing root. For all wing geometries in this work, a 0–45 deg pitch ramp motion with a nondimensional pitch rate of $K = \dot{\alpha}c/2U_\infty = 0.3$ is considered, Figure 9 shows the time variation of the angle of attack (which is the same as the pitch angle in this work) for this value of K. The equation for the pitch ramp motion is from Granlund et al. [30].

C. CFD

North Carolina State University's REACTMB-INS solver is used for the computational fluid dynamics calculations performed in this study. This finite volume solver formulates the time-dependent incompressible Navier–Stokes equations in an arbitrary Lagrangian/Eulerian fashion, thus enabling moving-mesh flow simulations on three-dimensional body-fitted computational meshes. An incompressible version of Edwards's low-diffusion flux splitting scheme (LDFSS) [93] is used for discretizing inviscid fluxes in space. Discretization of viscous terms is performed using a second-order central difference method. The LDFSS method is extended to higher-order spatial accuracy using the piecewise-parabolic method [94]. For time integration, second-order temporal accuracy is achieved by using an implicit artificial compressibility method [93] with subiterations performed at each physical time step for continuity equation convergence. A version of the Spalart–Allmaras one-equation eddy viscosity model, modified by Edwards and Chandra [95], is used for turbulence closure. REACTMB-INS has been shown to predict experimental trends accurately for pitching rectangular wings at similar conditions as used in this study [76]. The meshes used in this investigation were generated using Pointwise™ and contained about 35 million cells each. The meshes are clustered to the body surface with a minimum spacing of 5 μm and are generated so that nearly isotropic mesh resolution is maintained in the regions traversed by the LEV.

Table 1 Test cases

Case	Taper ratio	AR	Tip twist, deg	Sweep angle, deg	Airfoil
1	1	6	0	0	SD7003
2	1	6	10	0	SD7003
3	1	6	0	30	SD7003

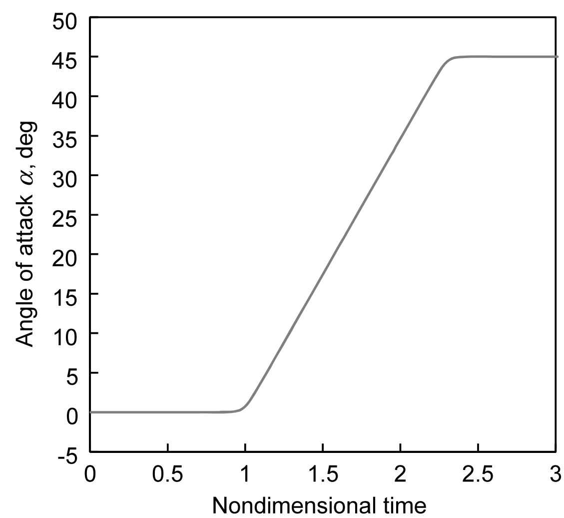


Fig. 9 Pitchup motion used in this study.

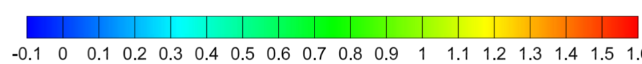


Fig. 10 Color bar for UVLM vortex-sheet color contours, colored using the vortex-ring strength Γ (in m^2/s).

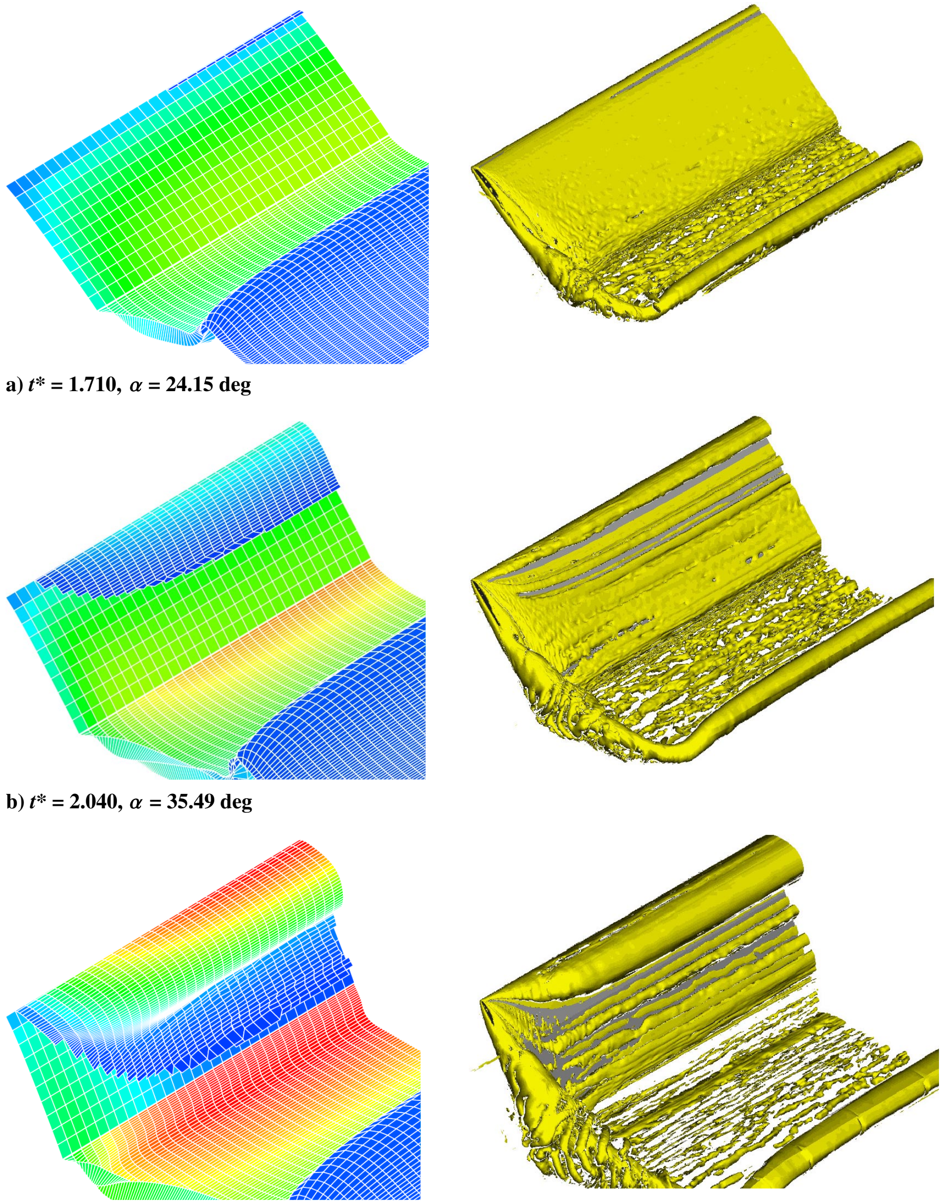
IV. Results and Discussion

The results of the UVLM with the new LEV sheet model for various finite-wing geometries are presented in this section. Predictions from the UVLM with the LEV model are compared with CFD results.

A. Development of LEV Structure

In this subsection, the shape of the LEV sheets and the distribution of vortex-sheet strength on the sheets, as predicted by the new LEV model, are presented and verified by comparing with results from CFD. The motion of the vortex sheets is determined by the local velocity field. Therefore, the shape of the sheet can be thought of as a surface swept by local streamlines, and the evolution of this surface can be compared with isosurfaces of vorticity-related quantities, such as the Q- riterion [96] or the vorticity magnitude itself. Although the iso-Q surface is not quantitatively equivalent to the vortex-sheet distribution, it allows for a qualitative comparison. The UVLM result shown is the surface representing the vortex sheet colored by the vortex-ring strength. The color key for the vortex-sheet strength in all the UVLM results is shown as a color bar in Fig. 10. The intent behind the use of color contours for the UVLM vortex sheets is to indicate the sheet-strength variation for a given time instant and between time instants in a motion. These colors do not have any correlation with the iso-Q surfaces from the CFD results.

Figure 11 shows the LEV development for case 1 for three representative time instants. Because the flow is symmetric, all pictures only show the left side of the wing and its flow structure. Figure 11a shows the UVLM and CFD results for a time instant immediately after LEV initiation. A close examination of the UVLM result shows a nascent LEV sheet at the wing root, indicating the spanwise onset of LEV initiation at that location, which compares well with the CFD observation. Shortly after LEV initiation, at t^* of 2.040, the UVLM prediction in Fig. 11b shows that shedding of the



a) t^* **= 1.710,** α **= 24.15 deg**

b) t^* **= 2.040,** α **= 35.49 deg**

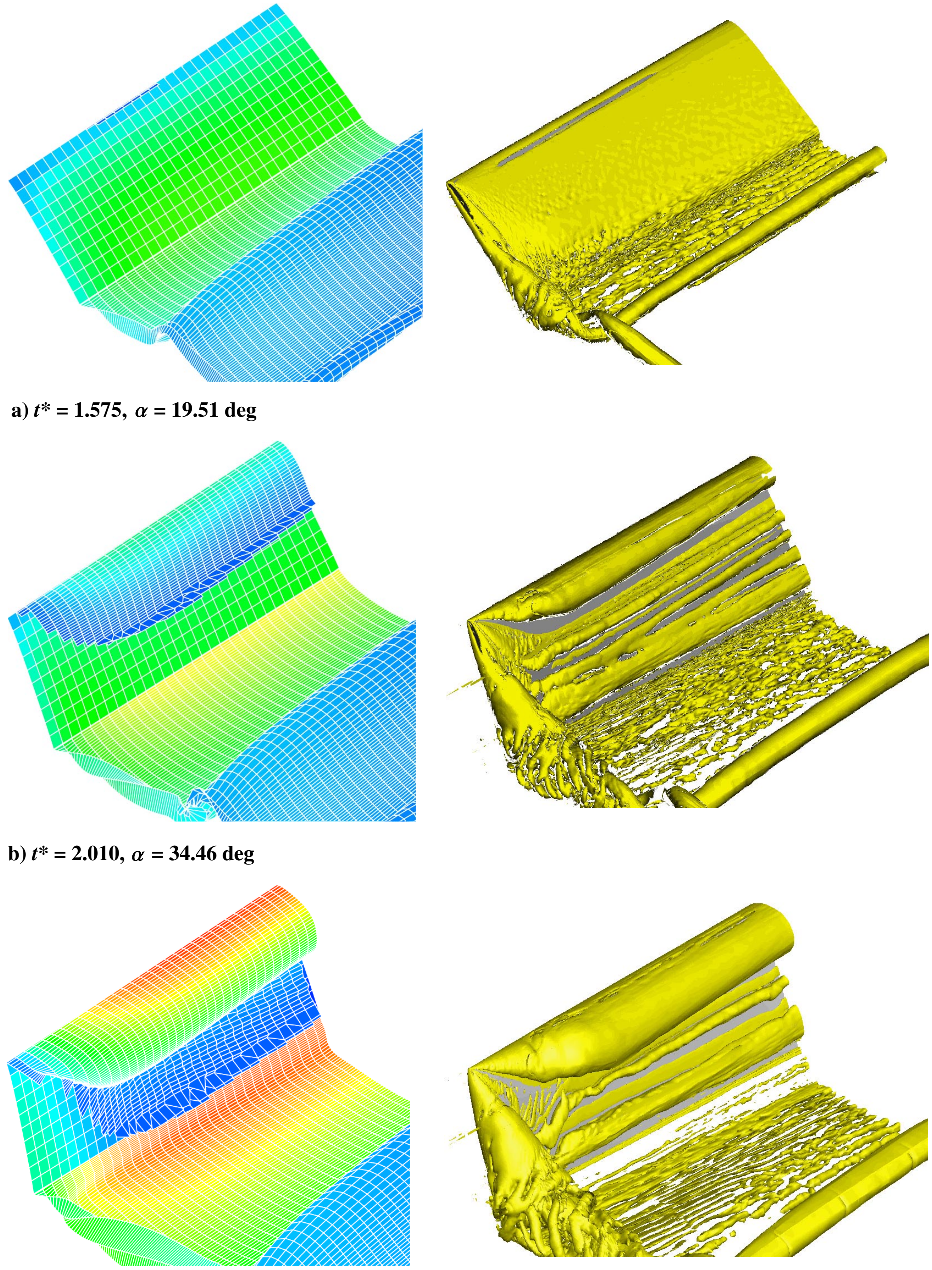
c) t^* **= 2.505,** α **= 45.00 deg**

Fig. 11 Comparison of LEV structure on the left side of the wing from UVLM (left) and CFD (right) for case 1.

LEV sheet has spread to a larger spanwise extent, which is largely in agreement with the CFD result, except that the UVLM appears to predict a slower spanwise movement of LEV formation than the CFD. One reason for the slower spanwise movement predicted by the UVLM is that the UVLM does not model a detached tip vortex over the tip chord, which could result in an error in the downwash in the outboard region of the wing. The rollup of the trailing-edge sheet predicted by the UVLM is seen to be in broad agreement with that seen in the CFD result. Figure 11c compares the UVLM and CFD results for a time instant shortly after the wing has reached the maximum pitch angle of 45 deg. The rollup of the LEV sheet predicted by the UVLM is seen to compare reasonably well with the shape of the rolled-up LEV structure in the CFD result. Overall, the comparison of the UVLM prediction of the vortex-sheet shapes in Fig. 11 is seen to be in broad agreement with the CFD results for case 1.

Figure 12 compares the UVLM-predicted and CFD-observed LEV flow structure for case 2 (wing with tip twist). At $t^* = 1.575$, a close examination of the UVLM prediction shows a nascent LEV sheet forming at approximately halfway between the wing root and tip, which agrees reasonably well with the CFD result for this time instant. This spanwise location for the onset of LEV initiation can be explained by the shape of the spanwise distribution of the LESP, which was shown in an earlier study [76] to have a maximum around $y = 0.5(b/2)$ for the wing with tip twist. In contrast, the spanwise distribution of the LESP for case 1 has a peak at $y = 0$, which explains the onset of LEV formation at the wing root for that case. The evolutions of the LEV and TEV sheets for two subsequent time instants are shown in Figs. 12b and 12c. As seen from Fig. 12c, both the UVLM and CFD results show a bulge in the rolled-up LEV sheet near $y = 0.75(b/2)$ for this case, which was absent in case 1. This bulge is because of the higher incidence of the outboard sections, resulting in greater accumulation of vorticity at these spanwise locations due to earlier LEV initiation.

The last verification is for case 3: a swept wing. This case was selected to highlight the effects of the onset of LEV formation close to the wingtip resulting from a peak in the spanwise distribution of the LESP at an outboard location close to the tip [76]. This behavior is a consequence of a motion-induced upwash on the wingtip because of the pitchup about the pivot location at the quarter-chord of the wing root section. As was done for the earlier cases, Fig. 13 shows the LEV flow structure as predicted by the UVLM and CFD at three time instants. The onset of LEV formation on this swept wing occurs near



a) t^* **= 1.575,** α **= 19.51 deg**

b) t^* **= 2.010,** α **= 34.46 deg**

c) t^* **= 2.505,** α **= 45.00 deg**

Fig. 12 Comparison of LEV structure on the left side of the wing from UVLM (left) and CFD (right) for case 2.

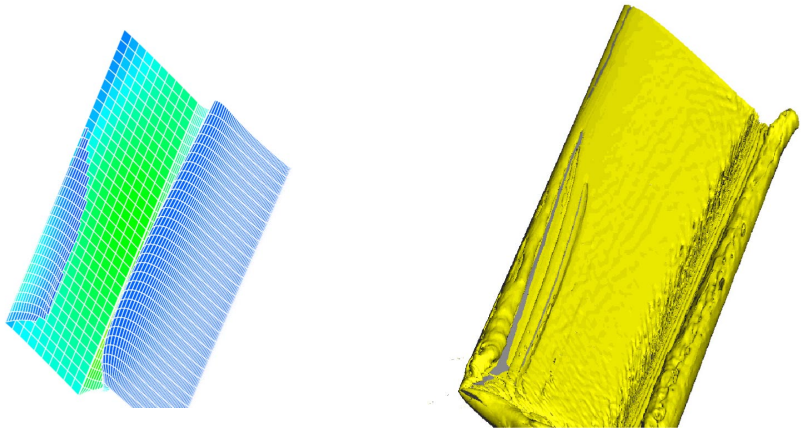
the wingtip, as seen from the nascent LEV sheet in the ULVM prediction and confirmed by the CFD result in Fig. 13a. The UVLM and CFD predictions for the two subsequent time instants in Figs. 13b and 13c show that, although the LEV sheet spreads in the spanwise direction toward the wing root, it is strongly tip dominated. The conical shape of the rolled-up LEV sheet is also reminiscent of the leading-edge vortex shedding from a delta wing.

Lastly, details of the inner structure of the UVLM prediction of LEV development are presented in Fig. 14, which shows a series of half-cut snapshots illustrating the LEV sheet distribution for the right side of the wing for case 1. The UVLM prediction successfully captures several stages of LEV formation: the onset of LEV formation from the wing root (Fig. 14a), the spanwise progression of the LEV initiation and initial rollup of the LEV sheet (Fig. 14b), further rollup of the LEV sheet (Fig. 14c), and the well-developed vortical structure on the wing upper surface (Fig. 14d). The UVLM prediction of the shape of the LEV sheet seems reasonable, given what is known from the literature: for example, from experimental observations [32] and tomography of the LEV structure [36].

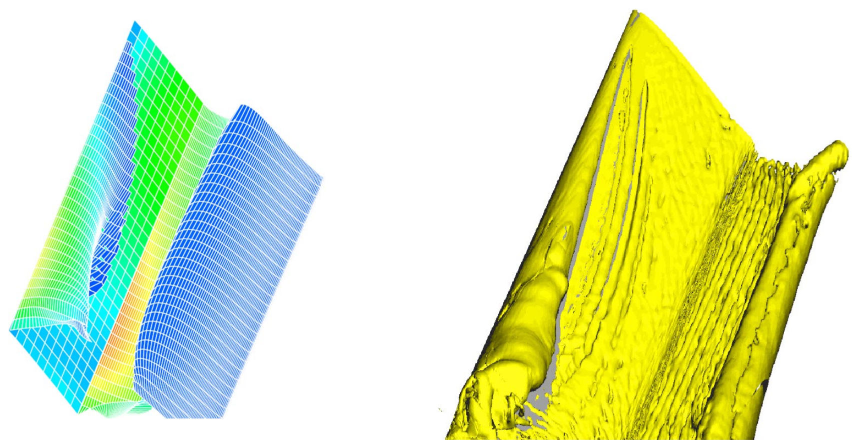
B. Prediction of Aerodynamic Loads with LEV

A comparison of the predicted lift coefficient C_L and pitching-moment coefficient about the wing root quarter-chord location C_M for case 1 is presented in Fig. 15. The UVLM results are also shown with the LEV formation turned off to show the predicted effects of the LEV formation. When the wing starts pitching at around $t^* = 1$, both the UVLM and CFD predict the effect of apparent mass with a positive spike in C_L and a negative spike in C_M. Likewise, when the wing stops pitching when α reaches 45 deg, there is a negative spike in C_L and a positive spike in C_M due to apparent mass effects. Although these spikes are present in both the UVLM and CFD results, the spikes in the UVLM predictions are larger than those in the CFD results. The reason for this overprediction by the UVLM is not clear.

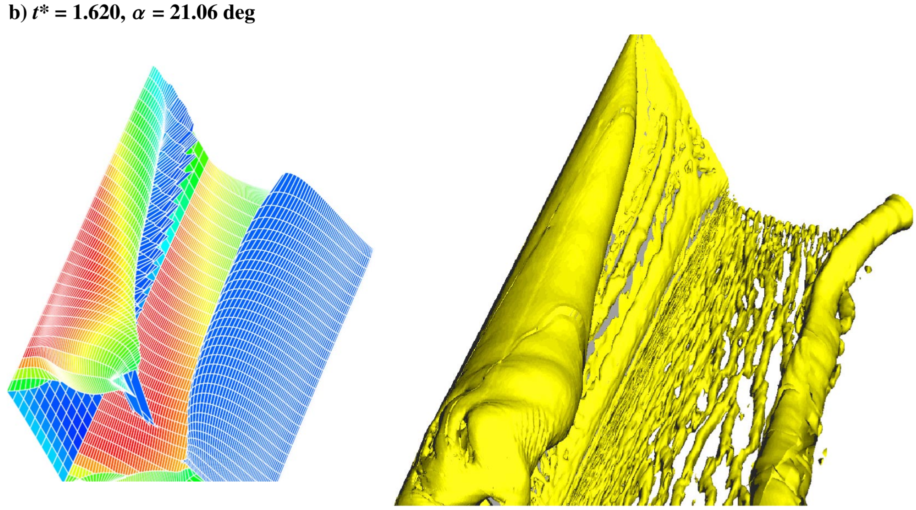
During the initial portion of the pitchup motion, C_L rises as the angle of attack α increases. In comparison to the CFD results, the slope of lift increase predicted by the UVLM is smaller, but the value of the predicted C_L is higher. The UVLM predicts a nearly constant value of C_M, whereas CFD predicts a less negative C_M with a negative



a) $t^* = 1.275$, $\alpha = 9.20$ deg



b) $t^* = 1.620$, $\alpha = 21.06$ deg



c) $t^* = 2.145$, $\alpha = 39.10$ deg

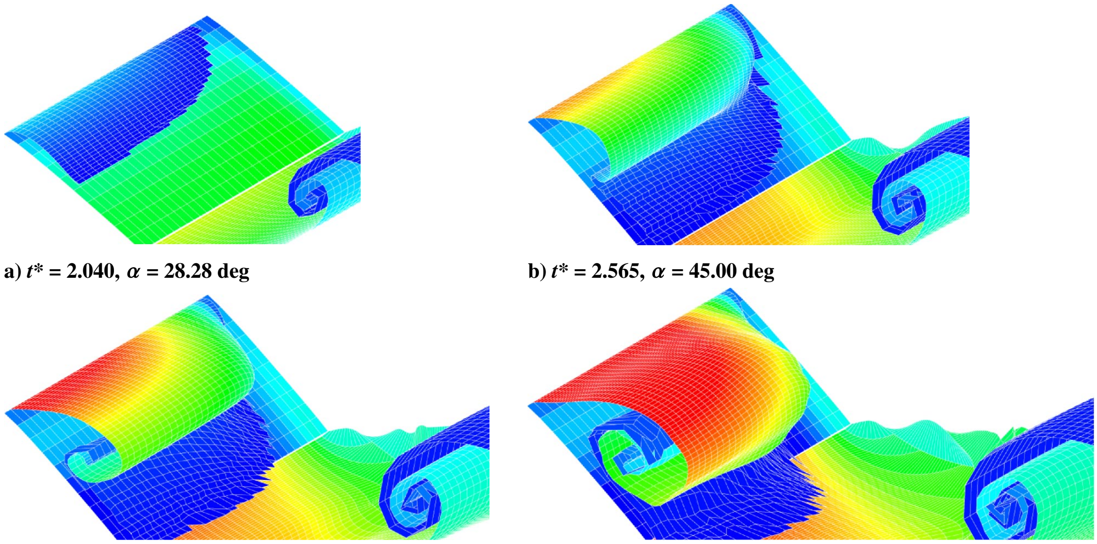
Fig. 13 Comparison of LEV structure on the left side of the wing from UVLM (left) and CFD (right) for case 3.

slope. The predictions for C_L and C_M from the UVLM and CFD agree well at the time instant when the UVLM predicts the onset of LEV formation.

Following the UVLM-predicted onset of LEV formation, the C_L with LEV formation exhibits a reduction in the rate of lift increase. In contrast, for the case in which LEV formation is switched off, the lift increase continues until α reaches the maximum value of 45 deg. During this period, the CFD-predicted C_L variation has a higher slope. Surprisingly, the UVLM agreement of the C_L prediction with CFD is a little better for the LEV-off case than when the LEV model is turned on. During this time interval between the predicted onset of LEV formation and the end of the pitchup motion, the prediction for C_M from the LEV-on UVLM simulation agrees excellently with the CFD prediction, whereas the LEV-off UVLM simulation predicts a slightly less negative C_M with a slight positive slope.

Following the end of the pitchup motion when the α is held at 45 deg, the LEV-off UVLM simulation shows constant C_L and C_M as expected. The predictions from the LEV-on UVLM simulation show smaller values for both C_L and C_M with negative slopes as compared to the LEV-off case.

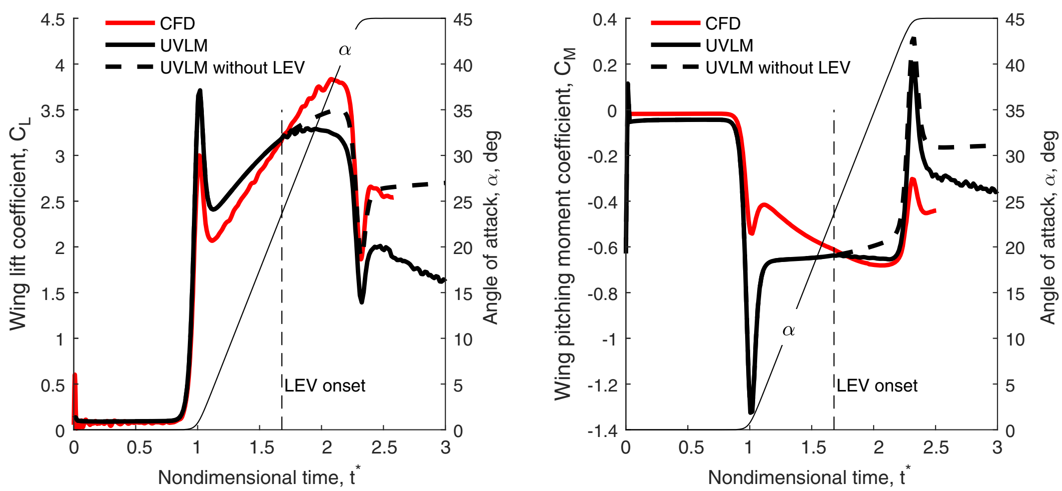
It is seen that, although some aspects of the force and moment predictions from the UVLM compare reasonably well with those from CFD, there are noticeable discrepancies. Although the reasons for the discrepancies are not fully understood, it is hypothesized that they may result from a combination of four possible causes. The first possible cause is the underestimation in the LEV-on UVLM



a) $t^* = 2.040$, $\alpha = 28.28$ deg b) $t^* = 2.565$, $\alpha = 45.00$ deg

c) $t^* = 2.820$, $\alpha = 45.00$ deg. d) $t^* = 3.300$, $\alpha = 45.00$ deg

Fig. 14 Time variation of the development of the LEV structure on the right side of the wing for case 1.



a) Comparison of C_L b) Comparison of C_M

Fig. 15 Comparison of C_L and C_M predictions from UVLM and CFD for case 1.

prediction of the leading-edge suction force F_S following LEV onset. This would result in a lower C_L as compared to CFD even though the C_M matches well. At the high angles of attack at and beyond the onset of LEV formation, an important contribution to C_L comes from the leading-edge suction. The second possible cause is the fact that the current UVLM does not simulate the influence of the detachment of tip vortices. Improvements in the tip-vortex model could potentially reduce the discrepancies in C_L. The third possible cause is the delay of the development of the LEV as predicted by the UVLM. As noted in the previous section, although the UVLM prediction shows qualitatively good agreement with the CFD observation, it is seen that the prediction of LEV development in the spanwise direction is somewhat slower than indicated in the CFD observation. This delay in the spanwise progression of LEV onset may cause an underestimation of vortex lift, and the weaker strength of the LEV in the UVLM prediction could contribute to the discrepancy in C_L. Finally, a fourth contribution to the discrepancies could be that the UVLM assumes an attached-flow model (except for vortex shedding at the leading edge) and does not model any trailing-edge flow separation.

Because the UVLM is a low-order method, it is significantly faster than higher-order methods like unsteady Reynolds-averaged Navier–Stokes (RANS) CFD calculations. The UVLM with the new LEV model requires about 25 min on a single processor of NCSU's high performance computing resource to complete a calculation like for case 1. For the same case, the RANS CFD calculation can take up to one day of runtime on 128 processors on the same resource. This computational speed shows the potential for the use of this UVLM in a preliminary design and analysis environment. Nevertheless, it needs to be mentioned that the computational cost of vortex methods typically increases steeply with increase in simulation time due to the need to calculate the mutual influence of the increasing number of vortex elements. Approaches such as deletion of vortex elements far from the surface (used, for example, in Ref. [97]) and amalgamation of vortices (used, for example, in Ref. [98]) have been used successfully in two-dimensional discrete-vortex simulations. These approaches are worth pursuing for future extensions of the current UVLM; however, implementation of these with three-dimensional free vortex sheets is considerably more complicated than in two-dimensional flows.

V. Conclusions

This paper presents the formulation of an unsteady vortex-lattice method augmented with a vortex-sheet model for representing the leading-edge vortex shedding along the span of a rounded leading-edge finite wing. The method uses criticality of the leading-edge suction to trigger initiation of leading-edge vortex (LEV) formation along the wing span, and it predicts results for the onset and nonuniform growth of the LEV sheet along the span that are in broad agreement with results from unsteady Reynolds-averaged Navier–Stokes (RANS) simulations. Following the onset of LEV formation on any strip of the wing, the shedding of a leading-edge vortex sheet formed of vortex rings is initiated from just upstream of the leading edge of the strip. To connect this sheet with the wing bound-vortex sheet, a special vortex ring, referred to here as a pseudovortex ring, is used. The method has been applied to pitching wings with differences in tip-twist and sweep angles. The shapes of the predicted LEV sheet structures for these wings qualitatively agree with the LEV structures seen from the iso-Q contours predicted by a RANS computational fluid dynamics (CFD) method. In particular, the unsteady vortex-lattice method (UVLM)-predicted differences in the spanwise locations for LEV onset and the nonuniform growth of the LEV sheets between the wings match those seen in the CFD results. Although some aspects of the force and moment predictions from the UVLM compare reasonably well with those from CFD, there are noticeable discrepancies in the prediction of the C_L variation, which may be attributed to contributions from incorrect prediction of leading-edge suction following the onset of LEV formation, the absence of a separated-tip-vortex model, a delay in the spanwise growth of the UVLM-predicted LEV sheet, and the attached-flow assumption that does not model any trailing-edge separation on the suction surface of the wing.

Although this new low-order formulation is much faster than unsteady RANS methods and is able to predict the onset and growth of the LEV sheet with reasonable accuracy, the current work is essentially a proof of concept of an LEV sheet model, the shedding of which is modulated by the spanwise variation of the leading-edge suction. There is a need for additional work along multiple avenues before the method can be routinely used for long-time simulations of practical vortex-dominated wing flows. One avenue is the use of higher-order vortex-sheet models and higher-order time stepping: both of which would make the method more robust for long-time simulations in which there is interaction between multiple sheets like those from the leading edge, trailing edge, and wingtips. A second avenue is to explore and incorporate the use of schemes such as vortex amalgamation and vortex deletion to avoid the nonlinear increase in computational cost for long-time simulations. The core contribution of the current work, which is the nonuniform spanwise shedding of the LEV sheet along a finite wing, is nevertheless applicable to advanced vortex models that improve on the current work. Thus, the current work serves as an important initial step in modeling unsteady flow past rounded leading-edge finite wings with leading-edge vortex shedding.

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